

PRML Assignment 1: Regression Experiment Report

Wang Hongyu (22371297)
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Abstract

This report presents a comprehensive study on regression methods applied to a given 2D dataset. We first implement three linear regression approaches: Ordinary Least Squares (OLS), Gradient Descent (GD), and Newton's Method. Since the data exhibits clear nonlinear patterns, we further explore polynomial regression with different degrees and Gaussian basis function regression. L2 regularization (Ridge Regression) is also investigated to address overfitting in high-degree polynomial models. The Gaussian RBF model achieves the best test MSE of 0.2868, significantly outperforming linear models (test MSE $\sim$ 0.595).

1. Introduction

Regression is a fundamental task in machine learning, aiming to learn a mapping from input features to continuous output values. In this assignment, we are given a 2D dataset (x, y) with 100 training samples and 100 test samples, where x is in $[0, 10]$. The task involves two parts:

- (1) Fit a linear model $y = w_0 + w_1x$ using three optimization methods (OLS, GD, Newton), and compare their training and test errors.
- (2) Since the data is nonlinear, explore more appropriate models and provide model selection rationale, experimental results, and analysis.

2. Methodology

2.1 Linear Regression

For a linear model $y = w_0 + w_1x$, we define the design matrix $\mathbf{X} = [1, x]$ and solve for the weight vector $\mathbf{w} = [w_0, w_1]^T$ that minimizes $\text{MSE} = (1/N) * \sum (y_i - x_i^T \mathbf{w})^2$.

OLS (Ordinary Least Squares): Closed-form solution $\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$. Provides the exact optimal solution in one step.

Gradient Descent (GD): Iterative optimization with update rule $\mathbf{w} \leftarrow \mathbf{w} - \text{lr} * \text{grad}(L)$, where $\text{grad}(L) = (2/N) \mathbf{X}^T (\mathbf{X}\mathbf{w} - \mathbf{y})$. We use learning rate $\text{lr} = 0.001$ for 5000 iterations.

Newton's Method: Second-order optimization using the Hessian matrix $\mathbf{H} = (2/N) \mathbf{X}^T \mathbf{X}$. Update rule: $\mathbf{w} \leftarrow \mathbf{w} - \mathbf{H}^{-1} \text{grad}(L)$. Converges in just 1-2 iterations for linear problems.

2.2 Nonlinear Models

Polynomial Regression: We extend the feature space using polynomial basis functions $\phi(x) = [1, x, x^2, \dots, x^M]$. The model becomes $y = \sum (w_j * x^j)$. Higher degree M captures more

complex patterns but risks overfitting. We normalize x to zero-mean unit-variance before computing polynomial features to prevent numerical overflow.

Gaussian Basis Functions: We use $\phi_j(x) = \exp(-(x - \mu_j)^2 / (2\sigma^2))$ with $K=15$ basis functions uniformly distributed across $[0, 10]$ with $\sigma = 0.5$.

L2 Regularization (Ridge Regression): To combat overfitting in high-degree polynomials, we add a penalty term: $L = \text{MSE} + \lambda \|w\|^2$. The solution becomes $\mathbf{w} = (\Phi^T \Phi + \lambda \mathbf{I})^{-1} \Phi^T \mathbf{y}$.

3. Experimental Studies

3.1 Data Overview

The dataset consists of 100 training points and 100 test points. As shown in Figure 1, the data exhibits a clear nonlinear oscillation pattern with noise, making linear fitting inherently limited.

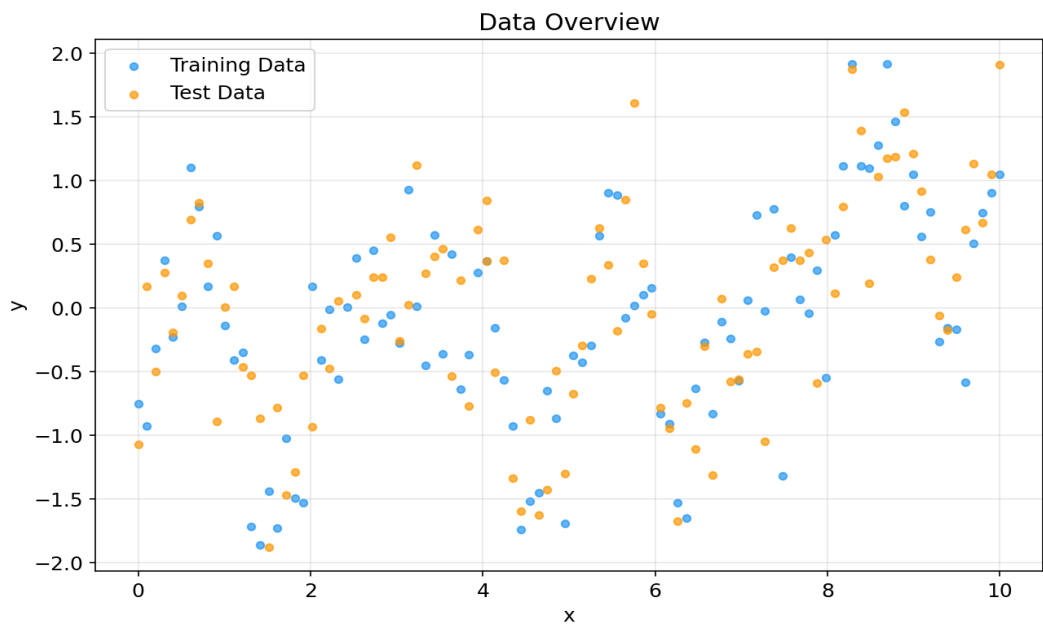


Figure 1: Training and test data scatter plot. The data shows nonlinear periodic patterns.

3.2 Linear Regression Results

Table 1 summarizes the results of three linear regression methods. All three converge to nearly identical solutions, confirming correctness. The test MSE (~ 0.595) is high, indicating poor linear fit.

Method	w0	w1	Train MSE	Test MSE
OLS	-0.6487	0.1089	0.6134	0.5950
GD	-0.6440	0.1080	0.6142	0.5934
Newton	-0.6487	0.1089	0.6134	0.5950

Table 1: Linear regression results. All methods converge to similar solutions.

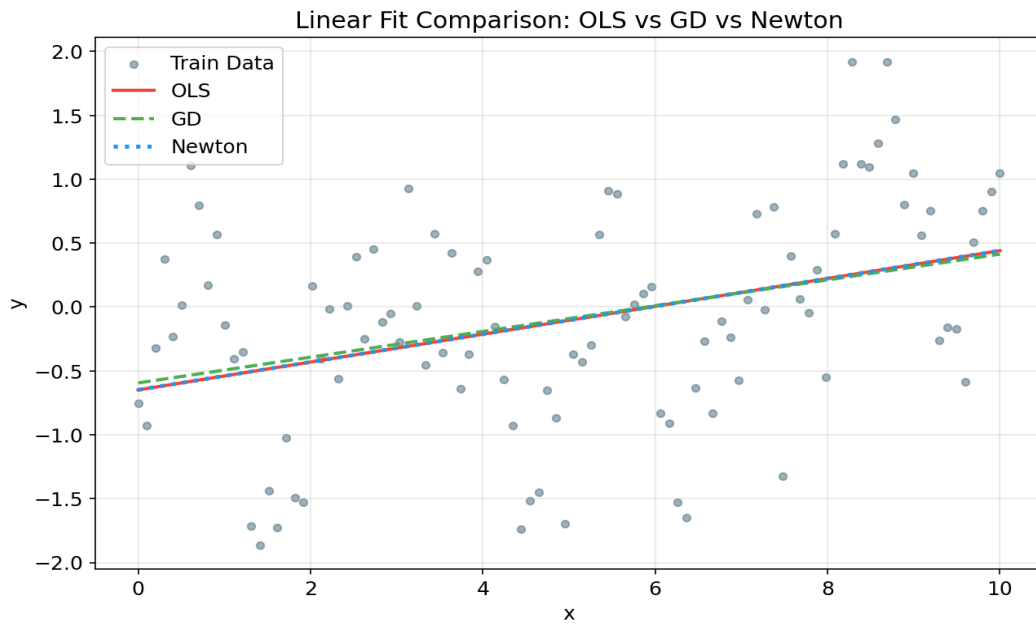


Figure 2: Comparison of three linear fitting methods. All produce nearly identical fits.

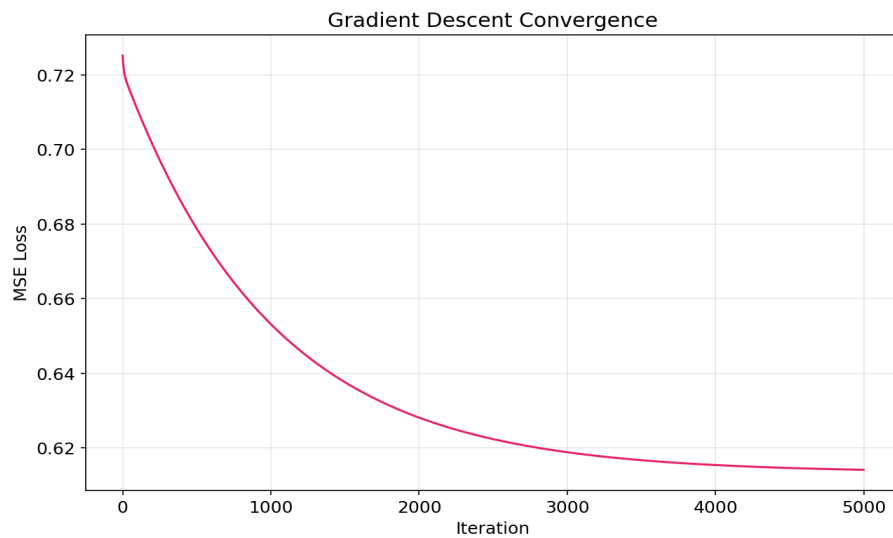


Figure 3: Gradient Descent convergence curve showing MSE decreasing over 5000 iterations.

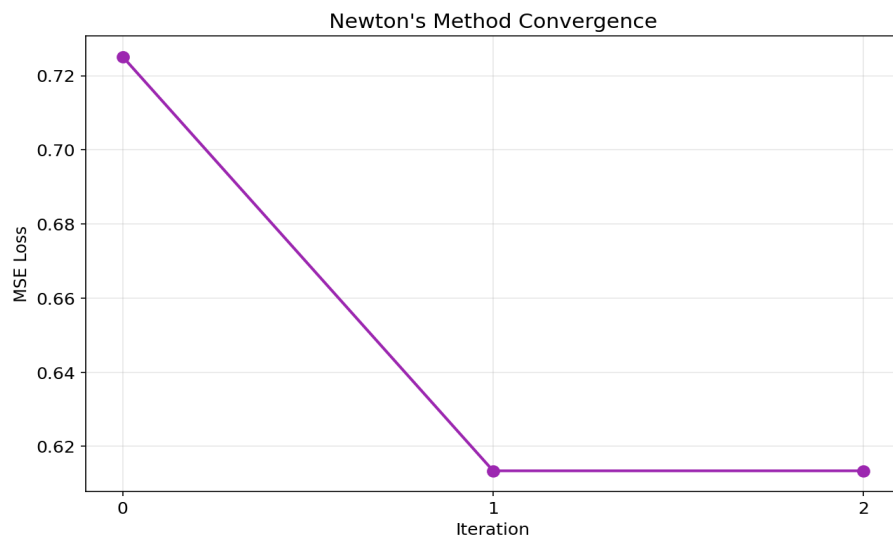


Figure 4: Newton's Method convergence - reaches optimum in just 2 iterations.

3.3 Polynomial Regression Results

Since the linear model is clearly insufficient, we apply polynomial regression with varying degrees. We normalize x before constructing polynomial features to ensure numerical stability for high degrees.

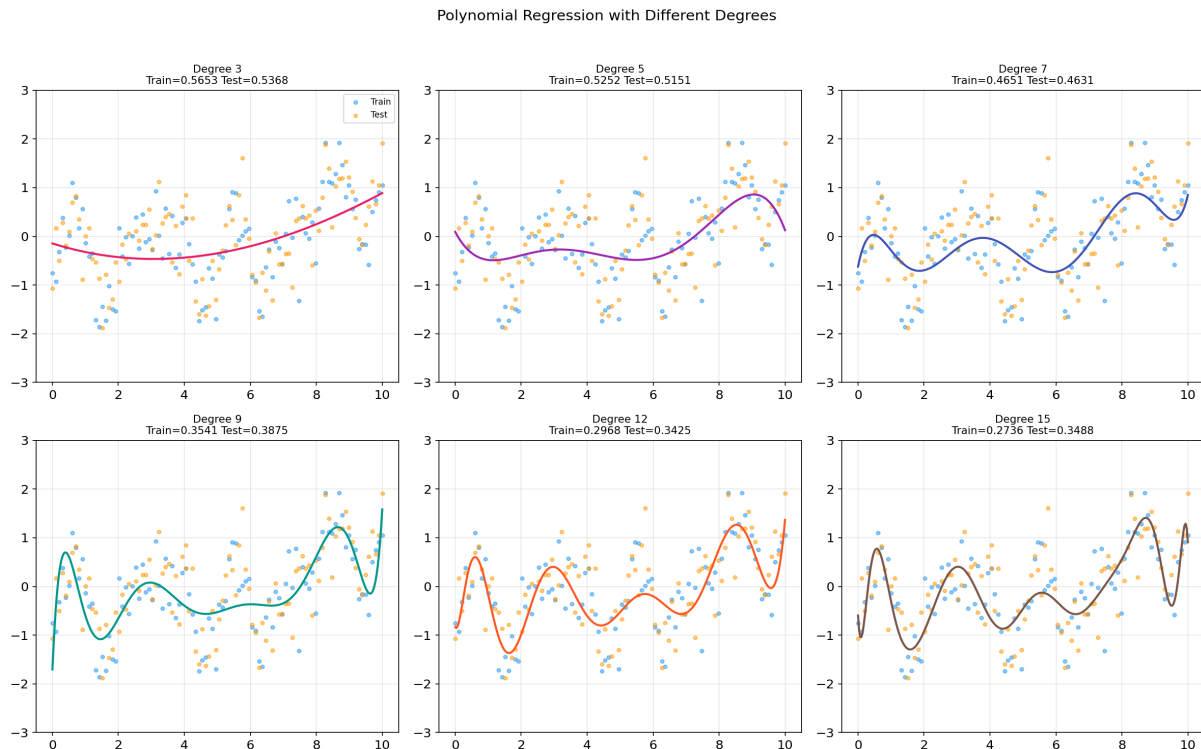


Figure 5: Polynomial fits with degrees $M = 3, 5, 7, 9, 12, 15$.

Degree M	Train MSE	Test MSE
3	0.5653	0.5368
5	0.5252	0.5151
7	0.4651	0.4631
9	0.3541	0.3875
12	0.2968	0.3425
15	0.2736	0.3488

Table 2: Polynomial regression errors for different degrees.

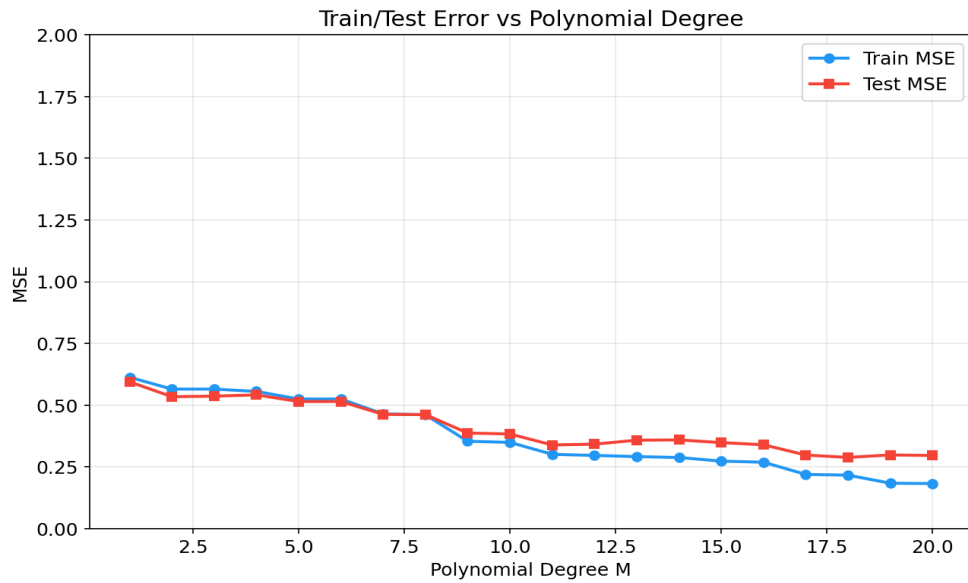


Figure 6: Train/Test MSE vs polynomial degree. The gap between curves shows the bias-variance tradeoff.

3.4 L2 Regularization (Ridge Regression)

For high-degree polynomials ($M=15$), we apply L2 regularization with different lambda values. As lambda increases, the model becomes smoother (lower variance but higher bias). lambda = 0.01 provides a good balance with Test MSE = 0.3807.

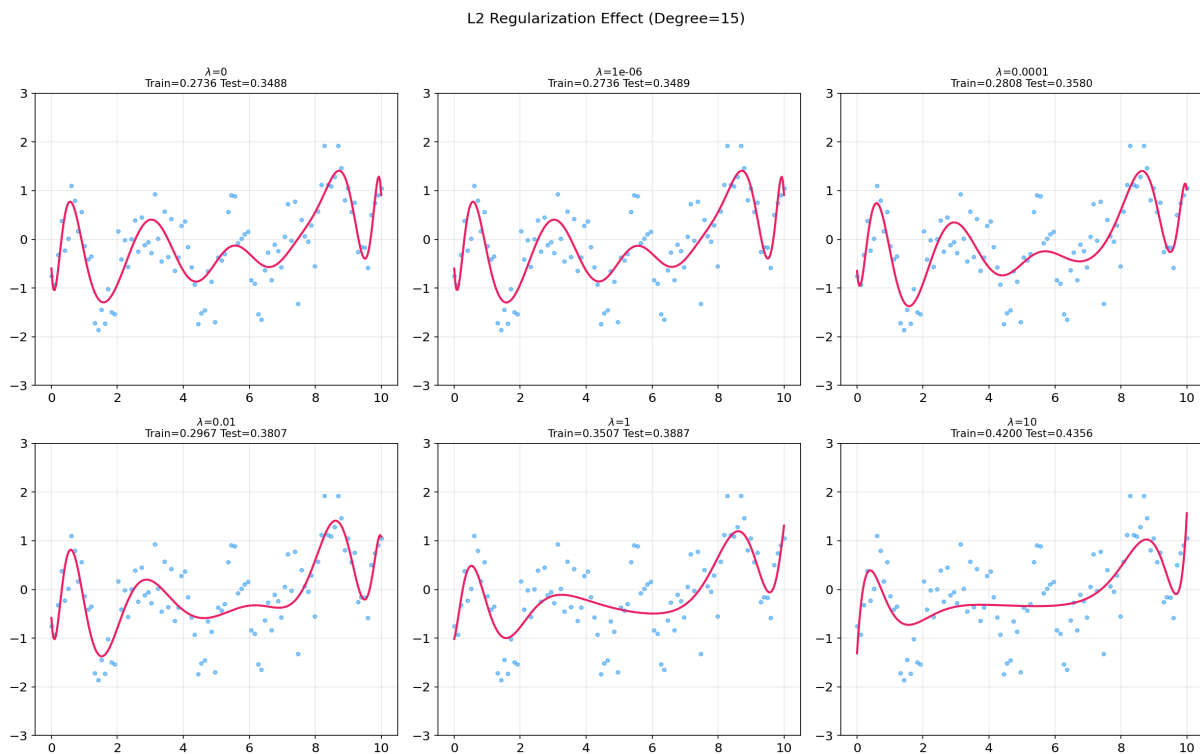


Figure 7: Effect of L2 regularization on degree-15 polynomial. Larger lambda progressively smooths the fit.

3.5 Gaussian Basis Function Regression

We use 15 Gaussian basis functions uniformly distributed in $[0, 10]$ with $\sigma = 0.5$, plus L2 regularization ($\lambda = 0.01$). This achieves the best overall performance with Train MSE = 0.1858 and Test MSE = 0.2868.

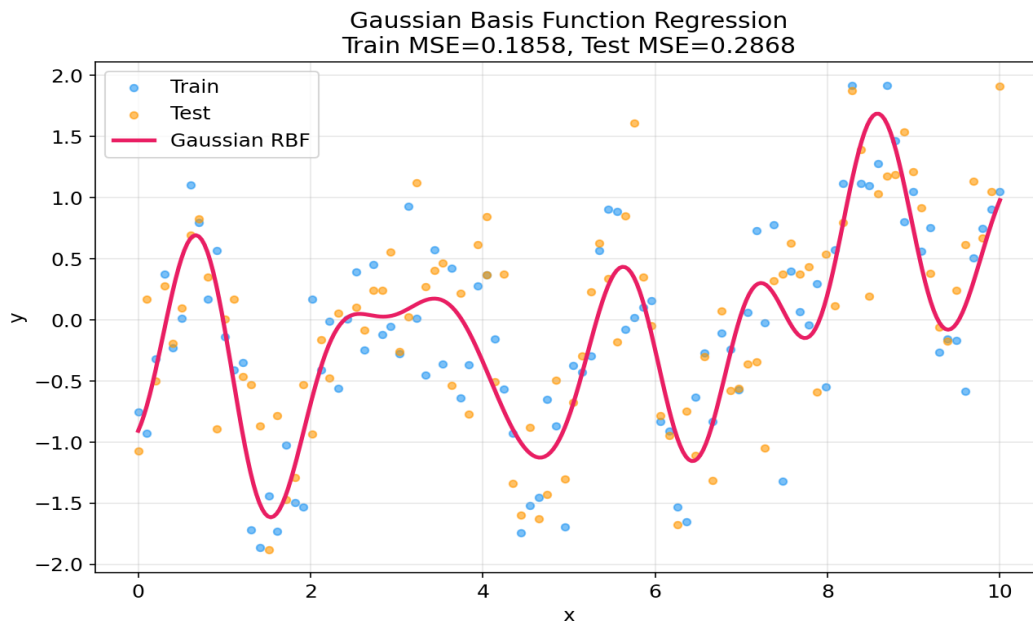


Figure 8: Gaussian RBF regression achieves the best fit among all methods.

3.6 Overall Comparison

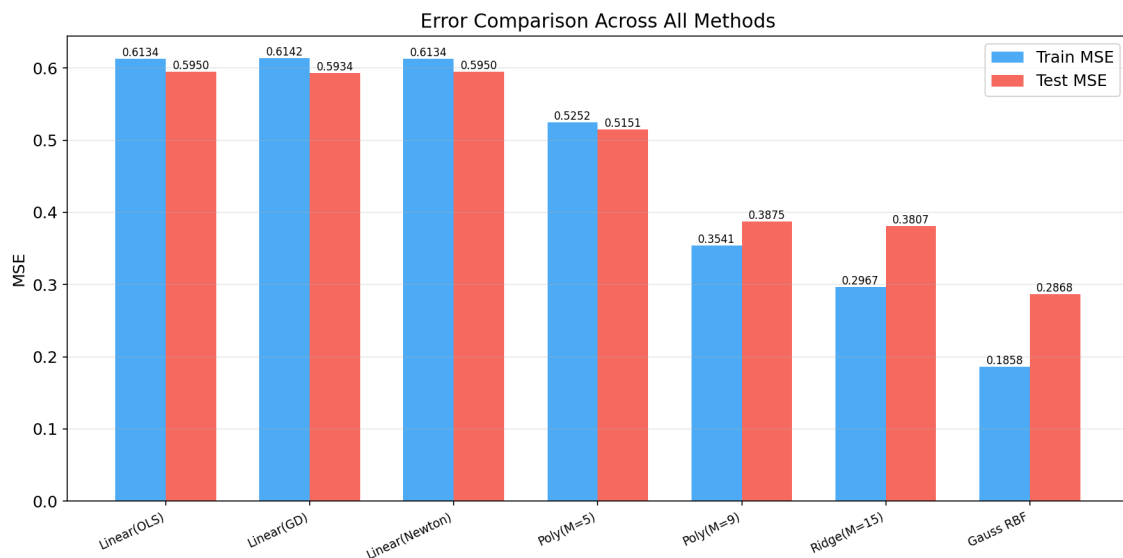


Figure 9: Error comparison across all methods. Gaussian RBF achieves the lowest test error.

4. Conclusions

This experiment demonstrates several key findings:

(1) All three linear optimization methods (OLS, GD, Newton) converge to the same solution, with Newton being the fastest (2 iterations). However, linear models are inadequate for this data (test MSE ~ 0.595).

(2) Polynomial regression significantly improves fit quality. Increasing degree reduces training error, but the test error gap demonstrates the classic bias-variance tradeoff. Degree $M=9$ offers a good balance without regularization.

(3) L2 regularization effectively controls overfitting in high-degree polynomials, allowing smoother fits with competitive test performance.

(4) Gaussian basis function regression achieves the best overall result (test MSE = 0.2868), suggesting that localized basis functions are well-suited for capturing the periodic oscillatory pattern in the data.

References

- [1] C.M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
- [2] T. Hastie, R. Tibshirani, J. Friedman, *The Elements of Statistical Learning*, 2nd ed., Springer, 2009.